Normal Probability Distributions

5.4 Sampling Distributions and The Central Limit Theorem

1. Use the Central Limit Theorem to find the mean and standard error of the mean of the indicated sampling distribution.
   a. **Height of Trees** The height of fully grown sugar maple trees are normally distributed with a mean of 87.5 feet and a standard deviation of 6.25 feet. Random samples of size 12 are drawn from the population and the mean of each sample is determined.
   b. **Red Meat Consumed** The per capita consumption of red meat by people in the United States in a recent year was normally distributed, with a mean of 110 pounds and a standard deviation of 38.5 pounds. Random samples of size 20 are drawn from this population and the mean of each sample is determined.

2. **Plumber Salaries** The population mean annual salary for plumbers is \$46,700. A random sample of 42 plumbers is drawn from this population. What is the probability that the mean salary of the sample is less than \$44,000? Assume σ = \$5600.

3. **Gas Prices: New England** During certain week the mean price of gasoline in the New England region was \$2.818 per gallon. A random sample of 32 gas stations is drawn from this population. What is the probability that the mean price for the sample was between \$2.768 and \$2.918 that week? Assume σ = \$0.045.

4. **Heights of Women** The mean height of women is the United States (ages 20 – 29) is 64.1 inches. A random sample of 60 women in this age group is selected. What is the probability that the mean height for the sample is greater than 66 inches? Assume σ = 2.71 inches.

5.5 Normal Approximations to Binomial Distributions

1. **Blood Type O^-** Seven percent of people in the United States have type O^- blood. You randomly select 30 people in the United States and ask them if their blood type is O^-.
   a. Find the probability that exactly 10 people say they have O^- blood.
   b. Find the probability that at least 10 people say they have O- blood.
   c. Find the probability that fewer than 10 people say they have O^- blood.
   d. A blood drive would like to get at least five donors with O^- blood. There are 100 donors. What is the probability that there will not be enough O- blood donors?

2. **Public Transportation** Five percent of workers in the United States use public transportation to get to work. You randomly select 250 workers and ask them if they use public transportation to get to work.
   a. Find the probability that exactly 16 workers will say yes.
   b. Find the probability that at least 9 workers will say yes.
   c. Find the probability that fewer than 16 workers will say yes.
   d. A transit authority offers discount rates to companies that have at least 30 employees who use public transportation to get to work. There are 500 employees in a company. What is the probability that the company will not get the discount.

3. **Favorite Cookie** Fifty-two percent of adults say chocolate chip is their favorite cookie. You randomly select 40 adults and ask each if chocolate chip is his or her favorite cookie.
   a. Find the probability that at most 23 people say chocolate chip is their favorite cookie.
   b. Find the probability that at least 18 people say chocolate chip is their favorite cookie.
   c. Find the probability that more than 20 people say chocolate chip is their favorite cookie.
   d. A community bake sale has prepared 350 chocolate chip cookies. The bake sale attracts 650 customers, and they each buy their favorite cookies. What is the probability there will not be enough chocolate chip cookies?